2026 edition

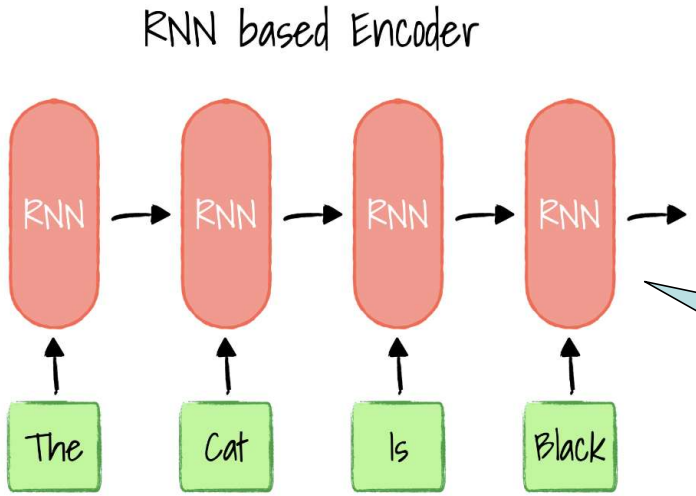
Deep Learning for Music Analysis and Generation

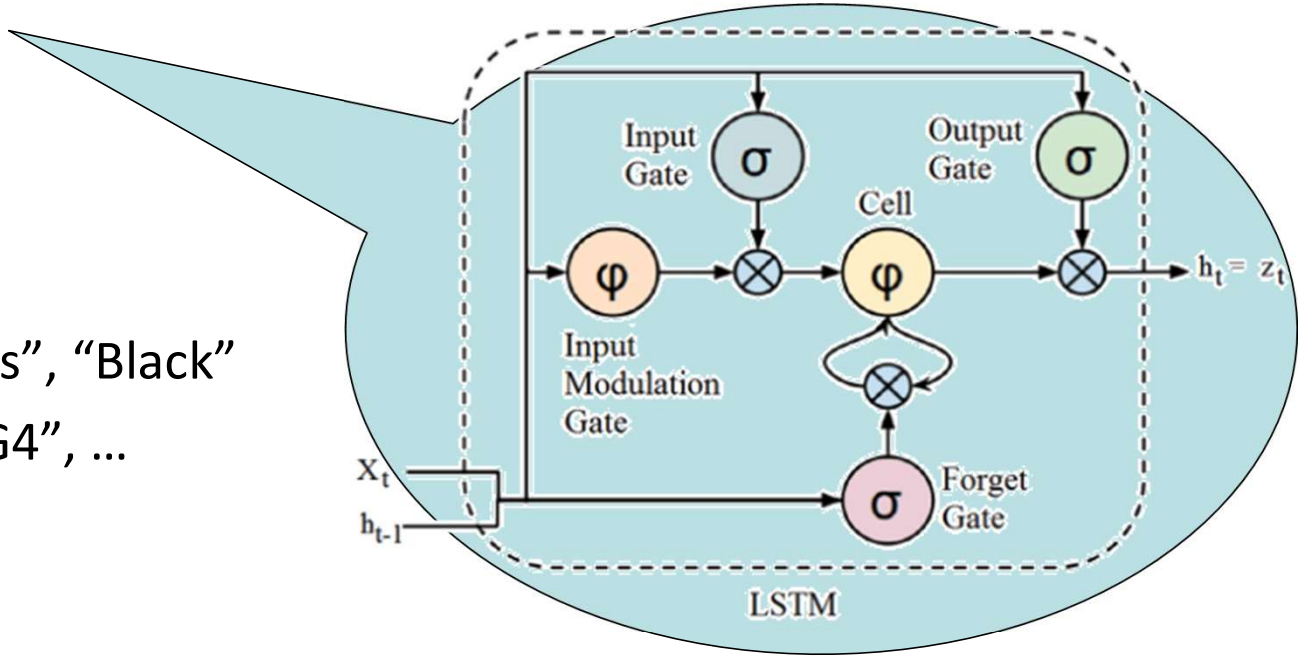
# Transformer



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yhyangtw@ntu.edu.tw

# RNNs

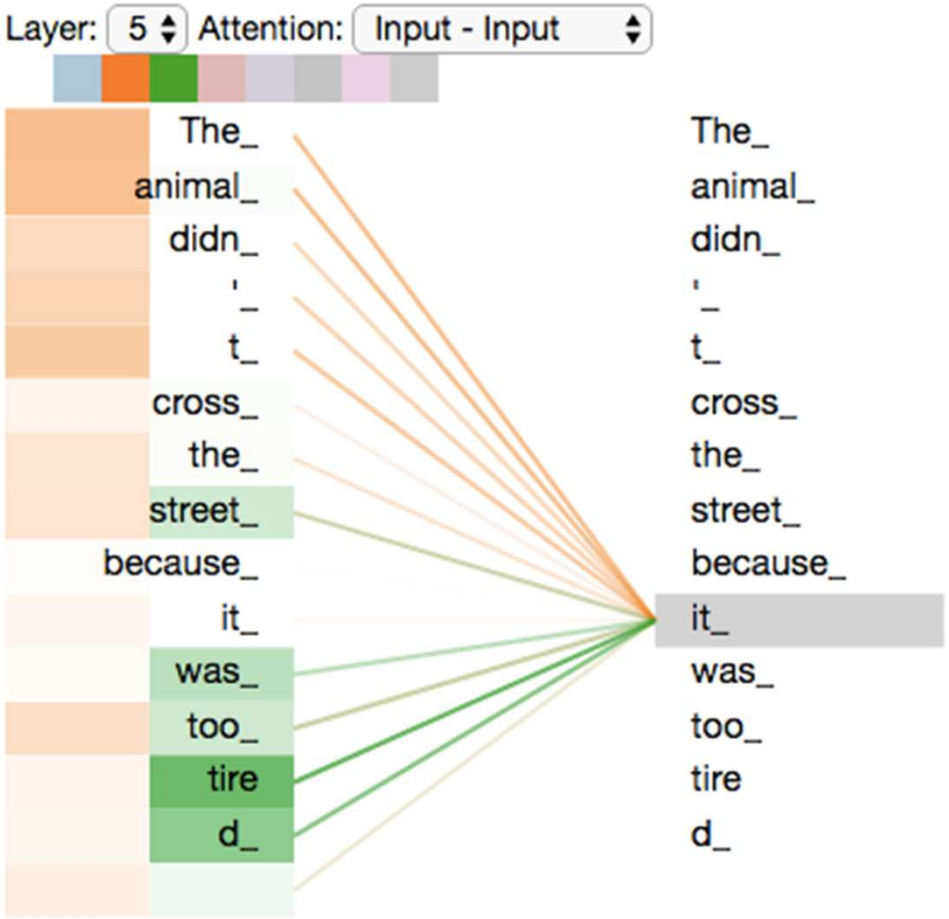


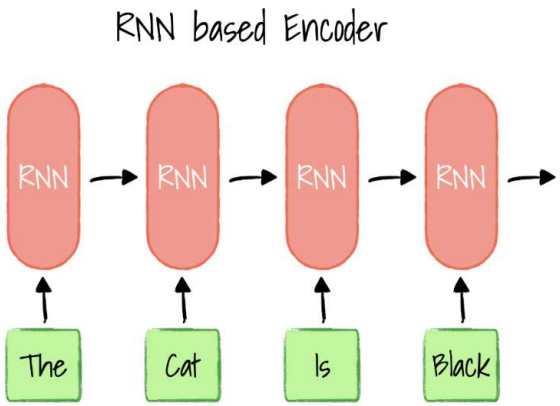

- Text tokens: “The”, “Cat”, “Is”, “Black”
- Music tokens: “C4”, “E4”, “G4”, …

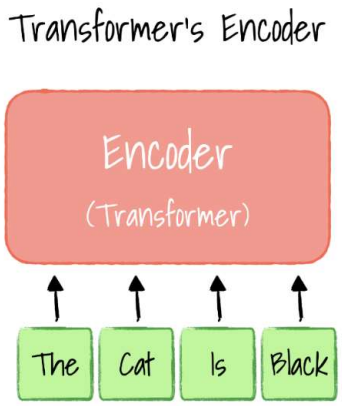
# Transformers

- Attention mechanism
  - Maintain a latent vector for ***every*** token in the history
  - Compute the correlation between the input with all the latents in the history
  - Need "**positional encoding**"
- In contrast, there is only **one** latent vector for the entire sequence in RNNs



# RNNs vs. Transformers




- **RNNs**
  - use only a latent vector
  - $y_t = f(y_{t-1}, h_{t-1})$

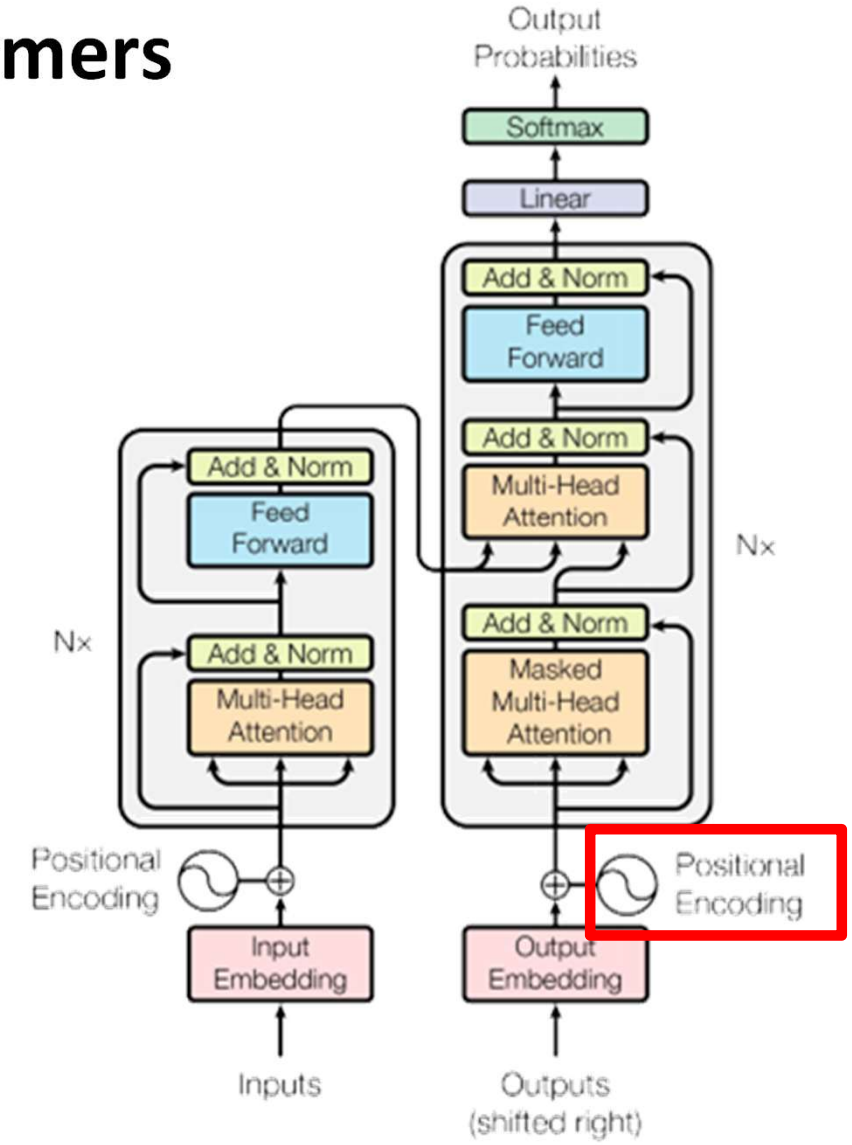
current output — last output — latent representation of the "history"

- **Transformers**
  - multiple latent vectors
  - $y_t = f(y_{t-1}, [h_{t-1}, h_{t-2}, \ldots, h_{t-L})$

latent representation of $t$-1 — latent representation of $t$-$L$

# Transformers

- There are encoders & decoders
  - Encoder: cross-attention
  - Decoder: self-attention
- **Decoder-only** architecture
  - Usually used for unconditional, or prompt-based generation
  - Mainly talk about this one today
- **Encoder/decoder** architecture
  - Usually used for sequence-to-sequence generation



# Transformer (from Prof. Jia-Bin Huang)

https://www.youtube.com/watch?v=rcWMRA9E5RI



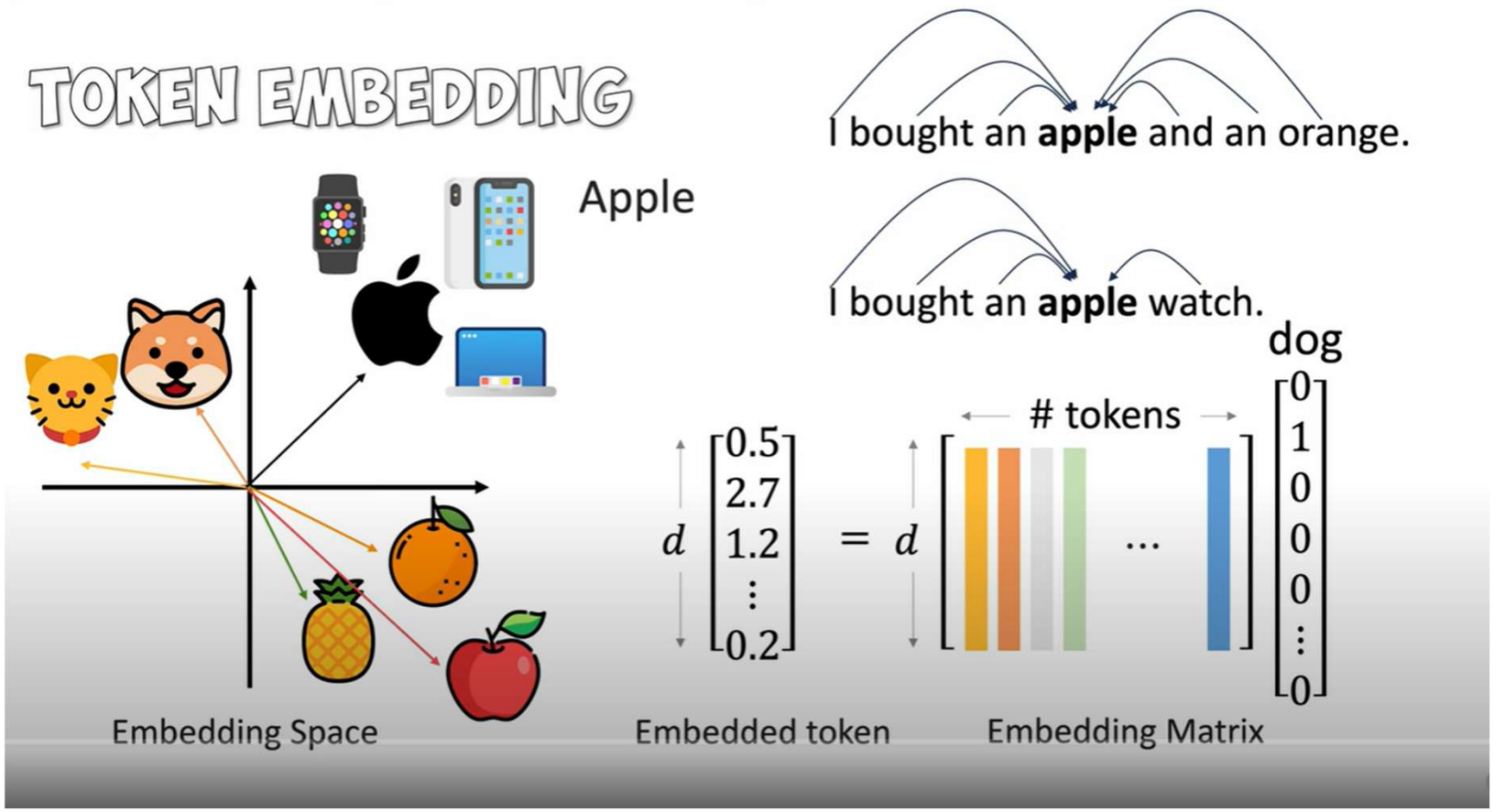

But What Are Transformers?

Jia-Bin Huang • 6.9K views

Transformers is arguably the most influential neural network architecture in the last decade, powering the current boom of...

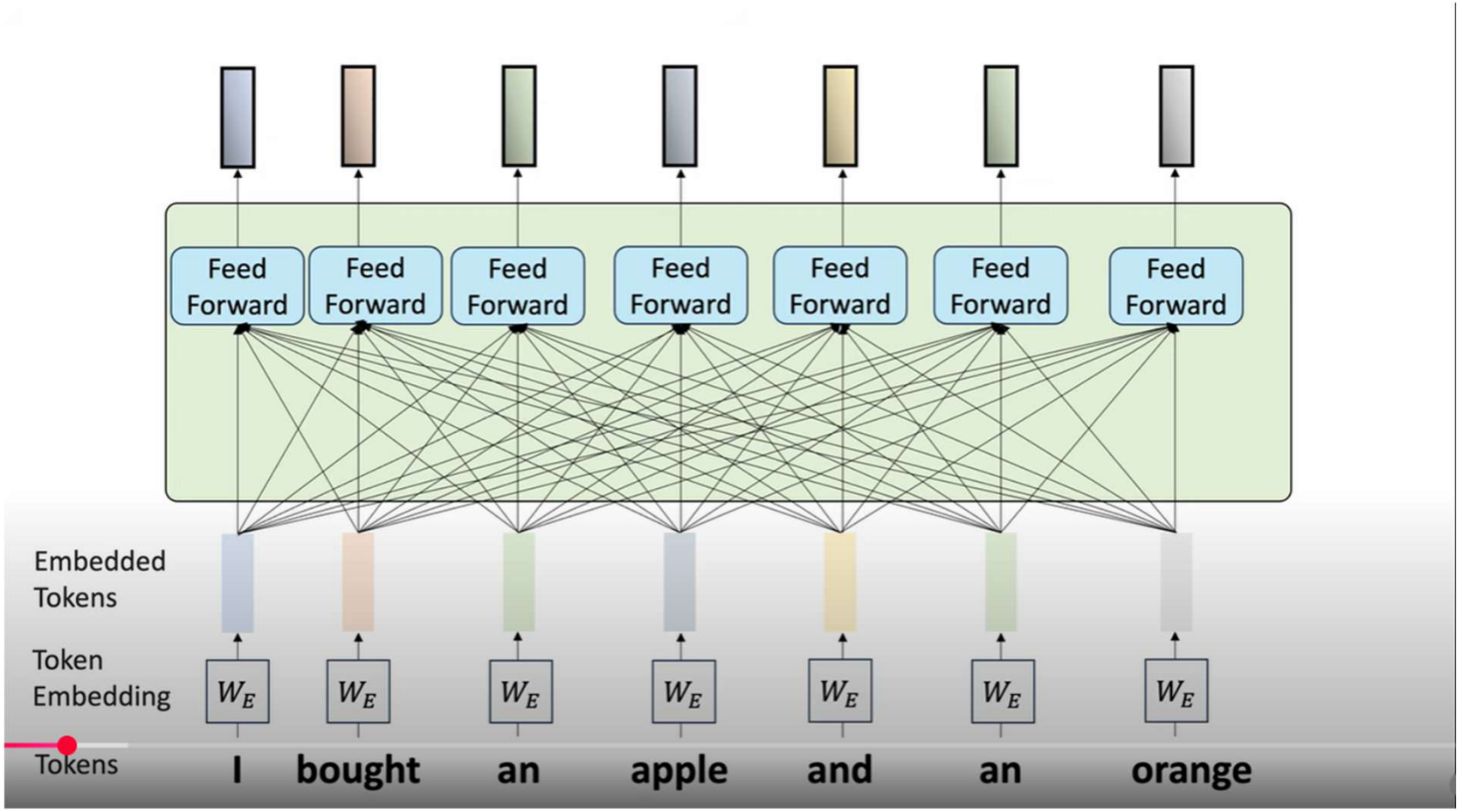
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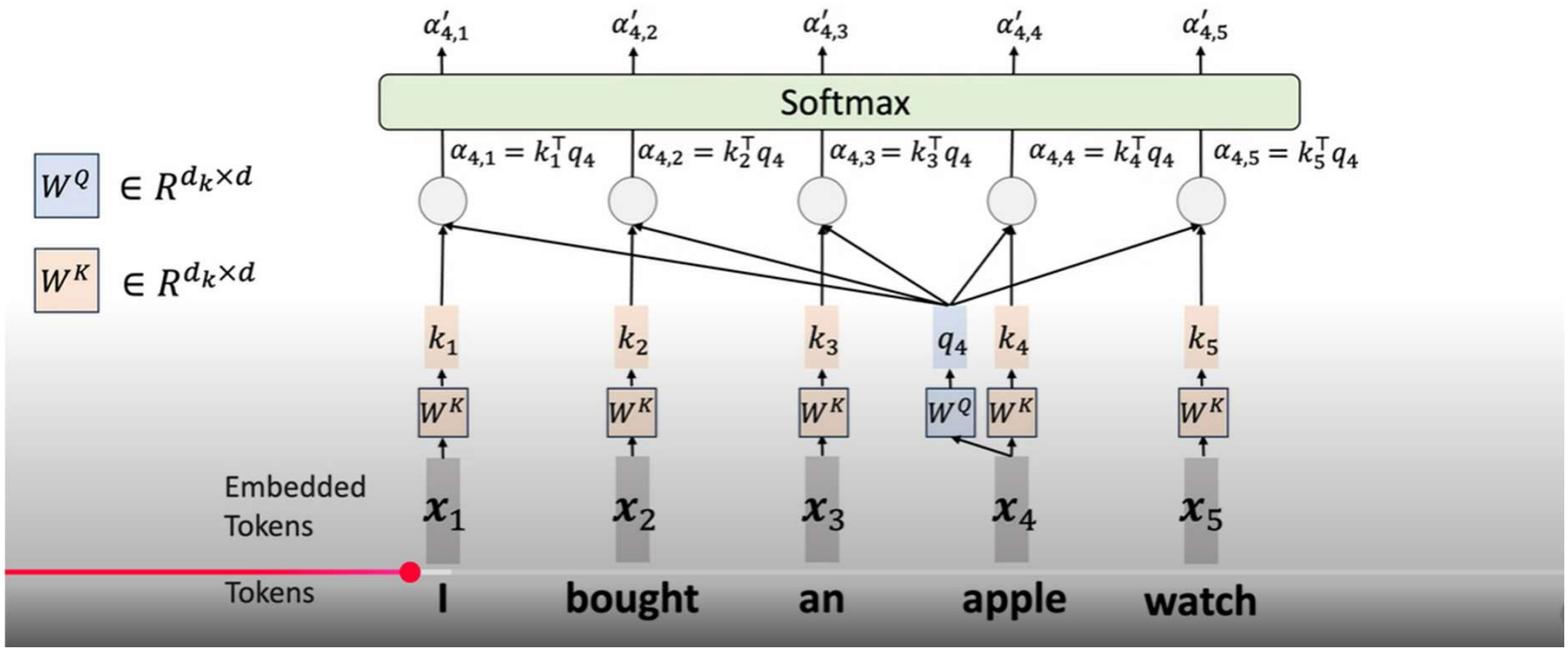
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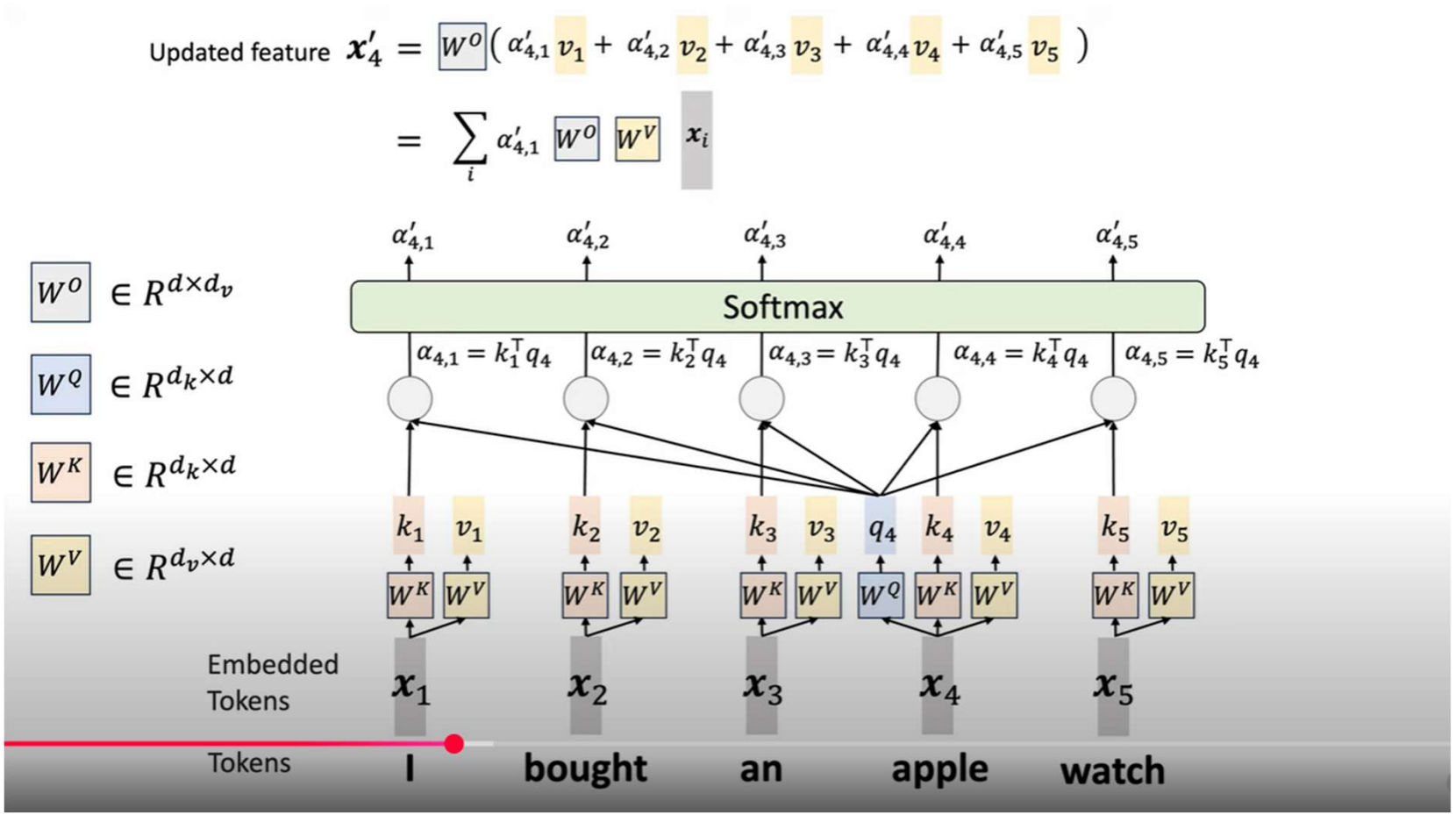
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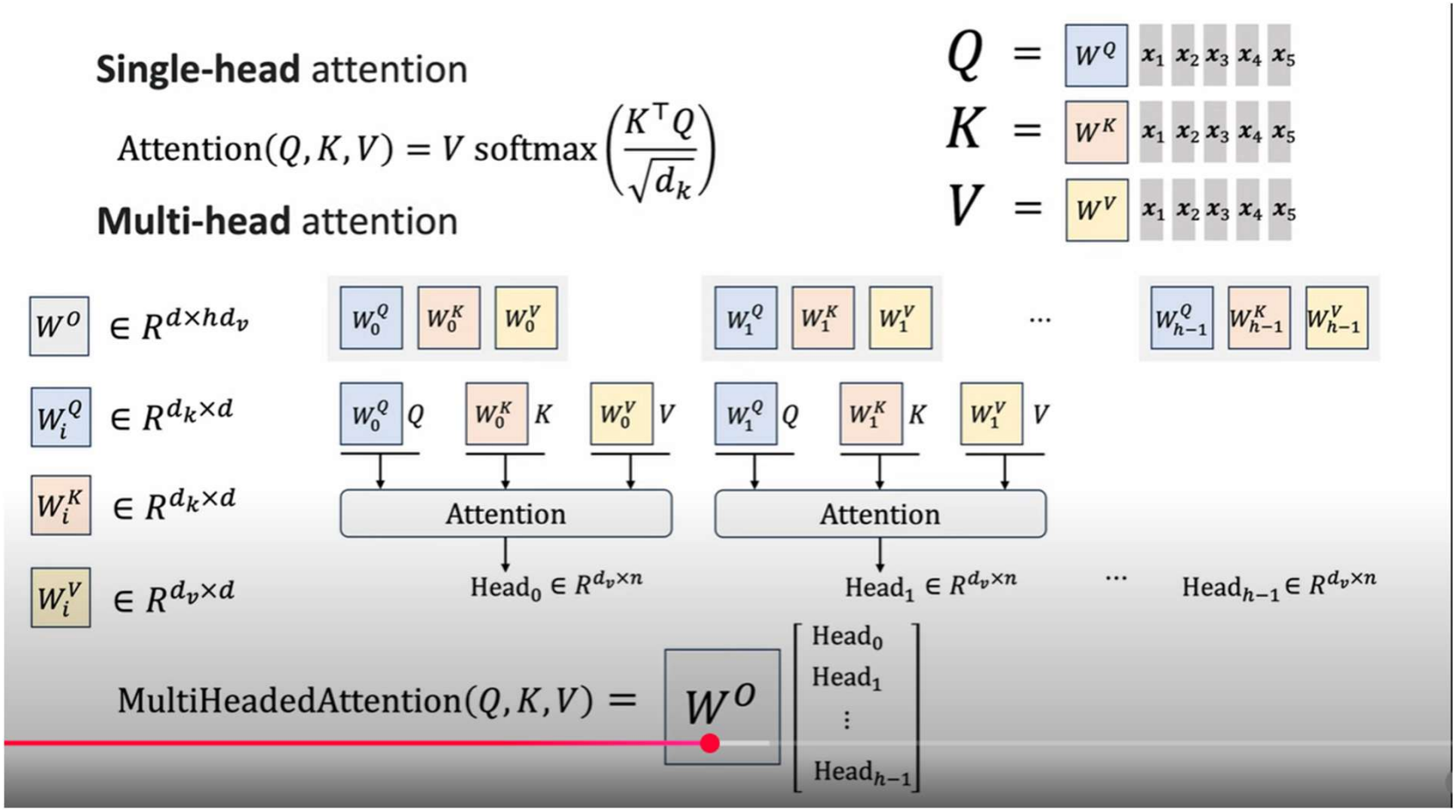
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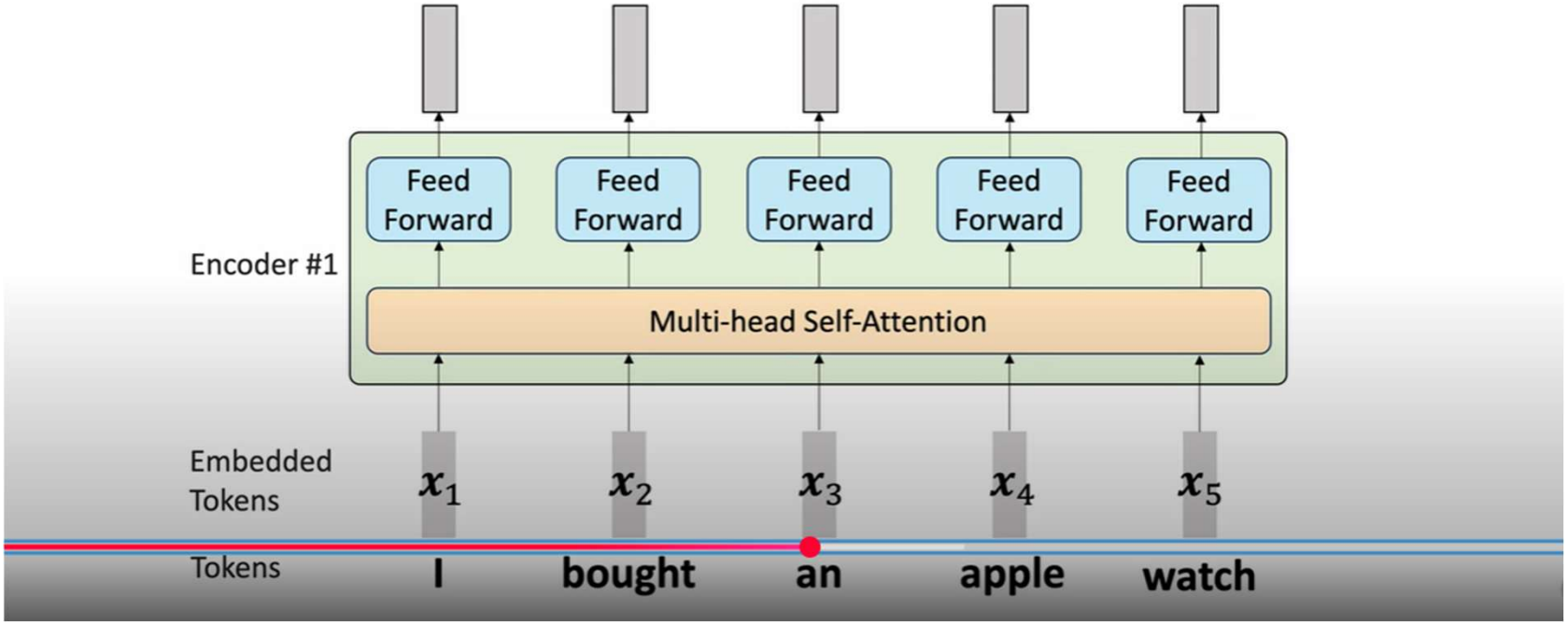


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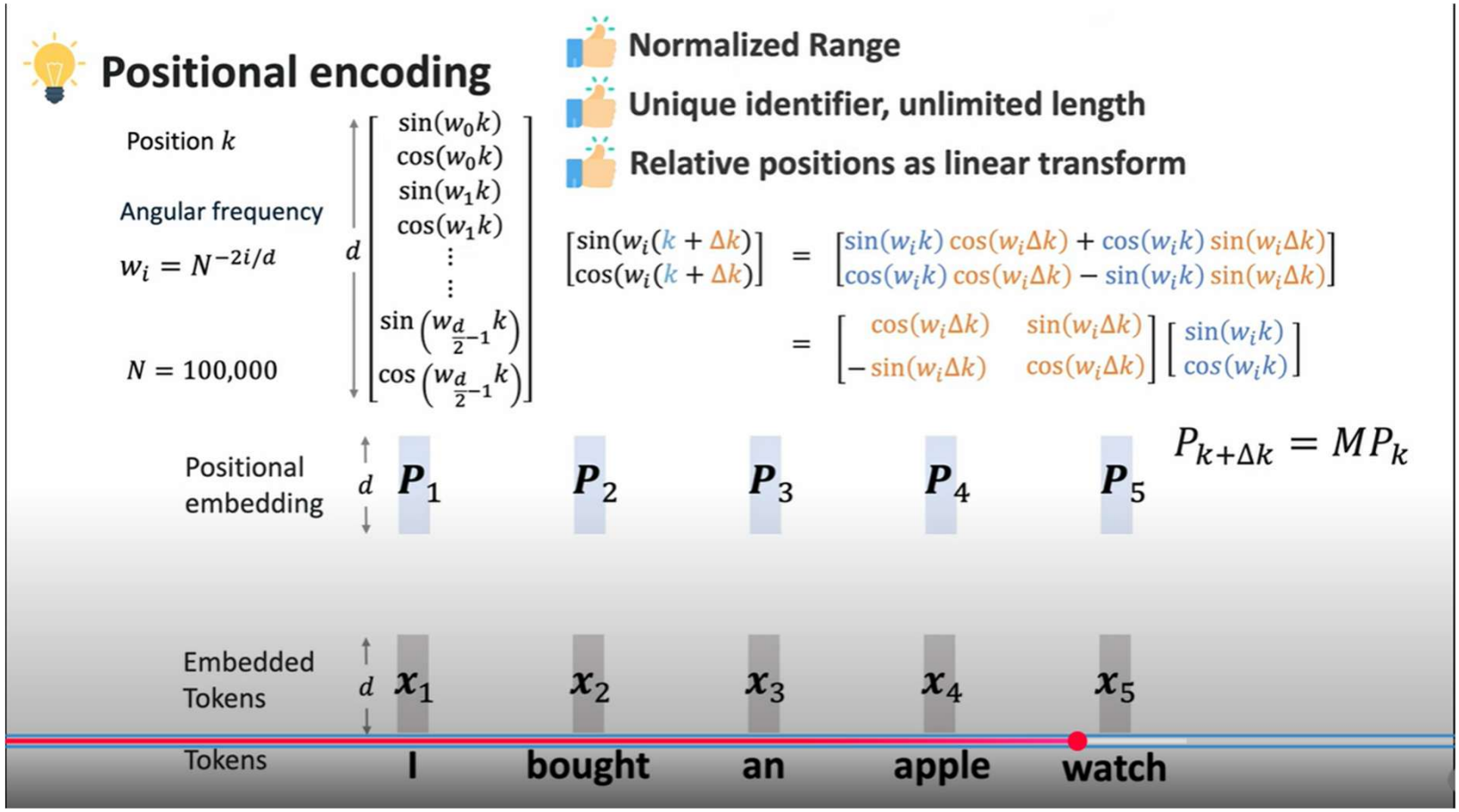
**Feed Forward Network (FFN)**

$$FFN(\boldsymbol{x}) = \boldsymbol{W}_2 \mathrm{ReLU}(\boldsymbol{W}_1 \boldsymbol{x} + \boldsymbol{b}_1) + \boldsymbol{b}_2$$



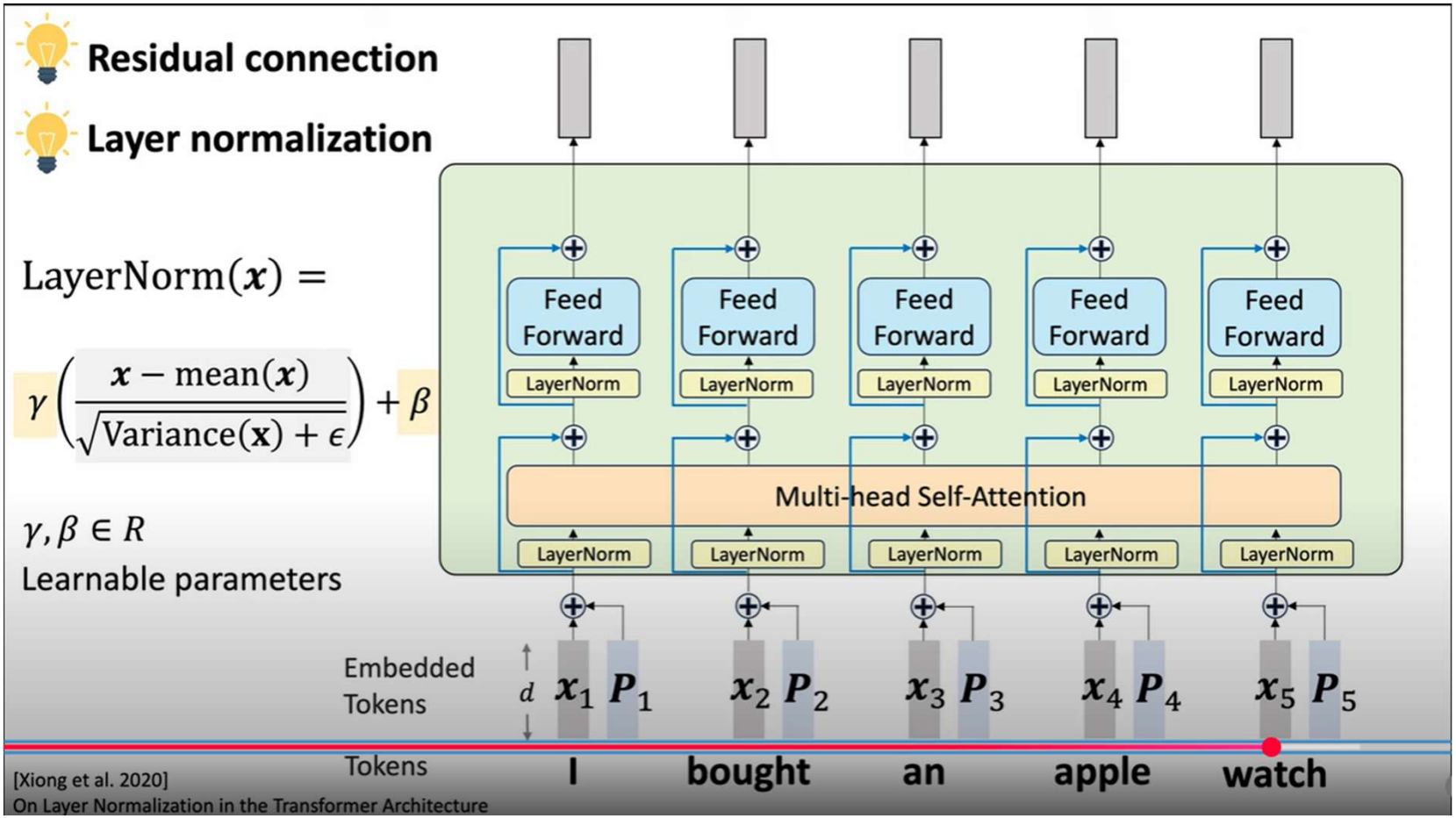
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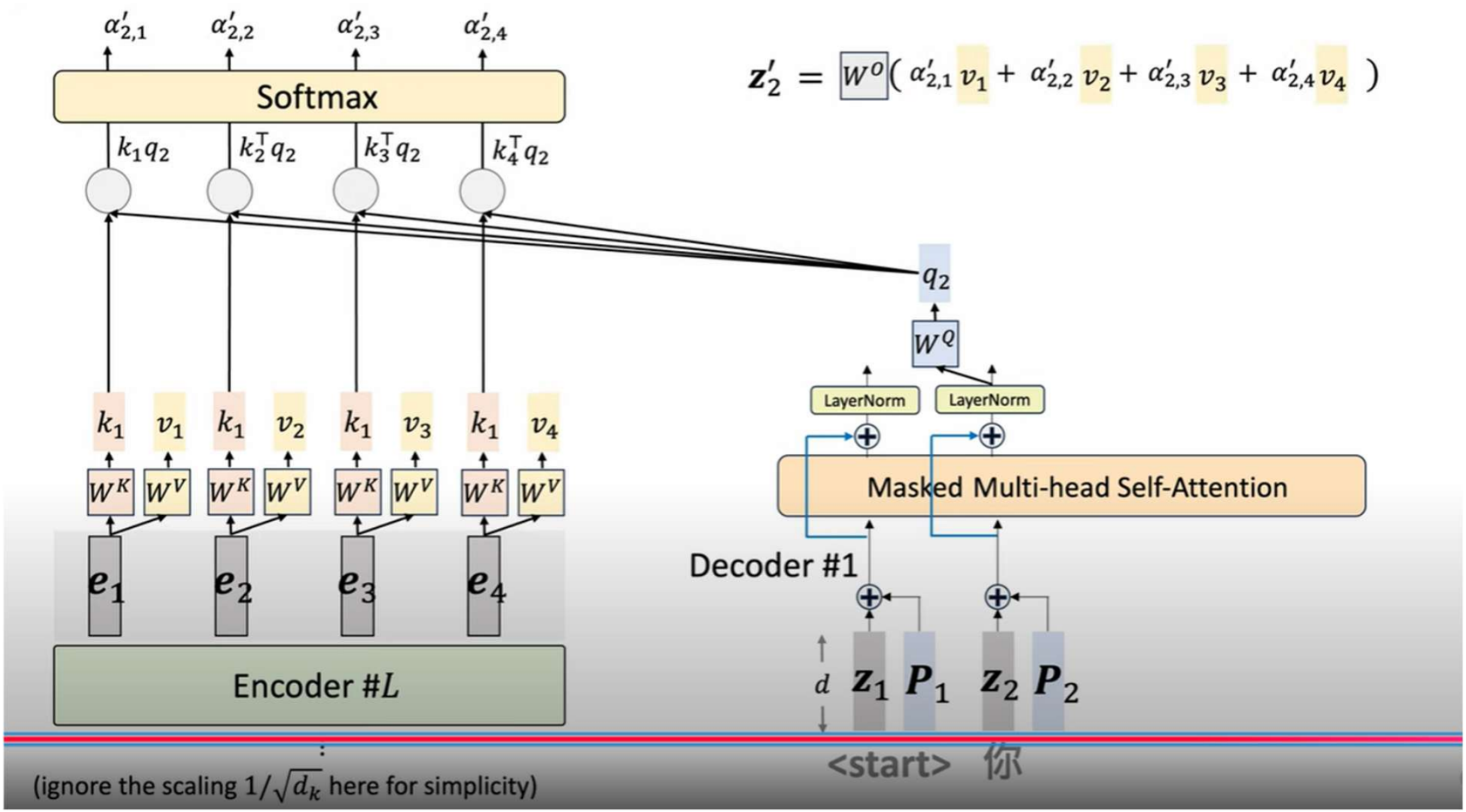
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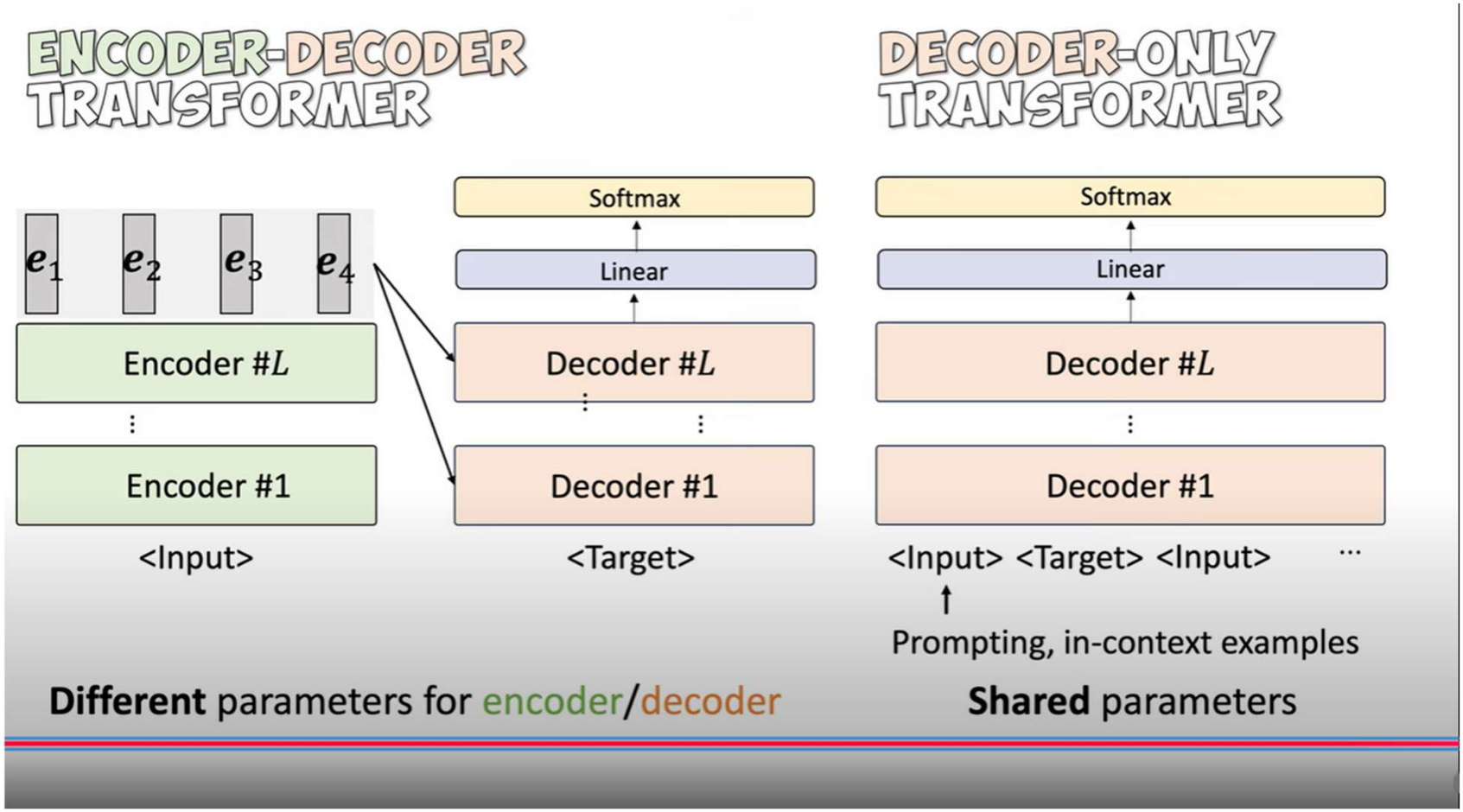
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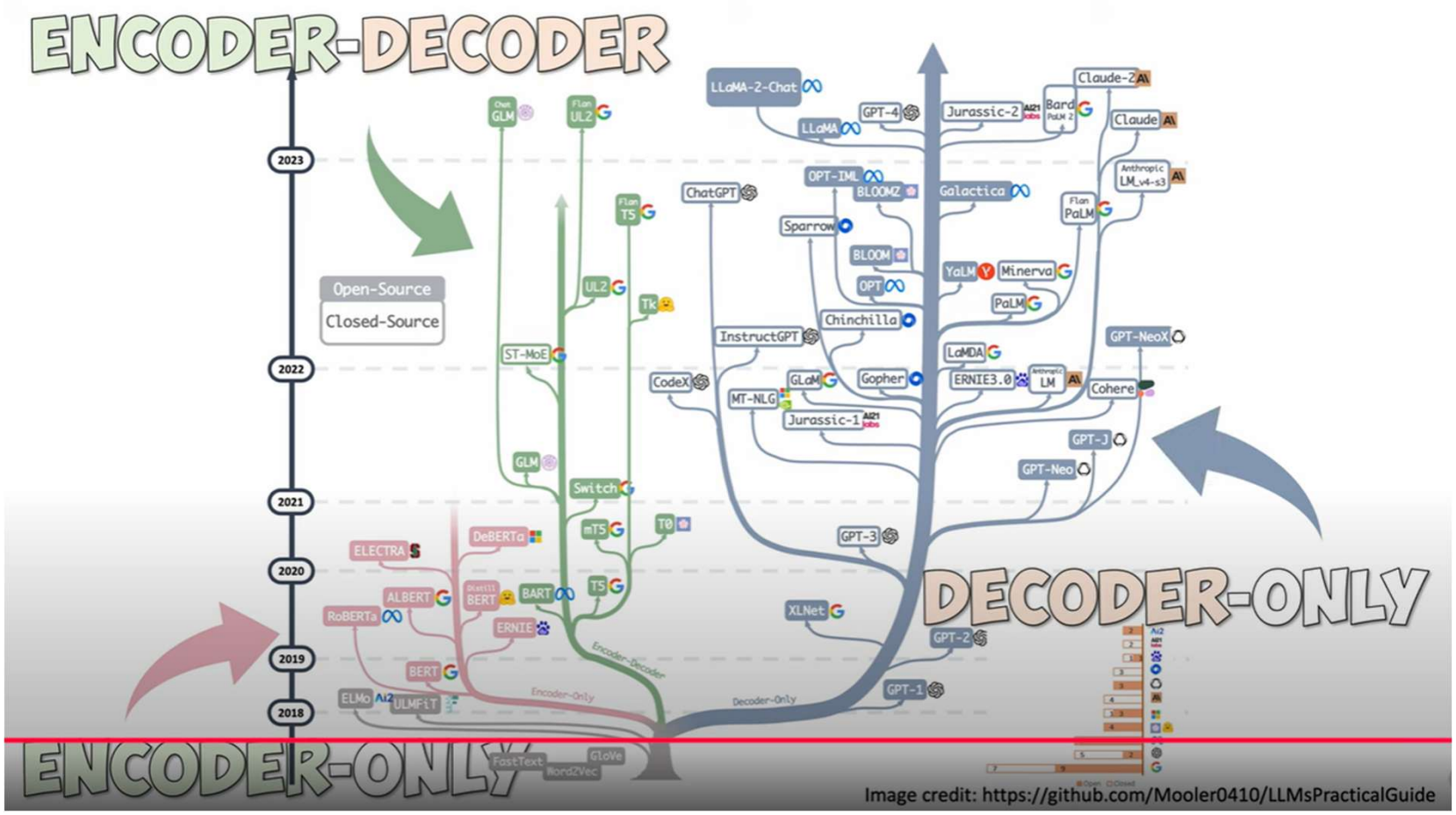
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# Transformer (from Prof. Jia-Bin Huang)

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# Positioning Embedding (from Prof. Jia-Bin Huang)

https://www.youtube.com/watch?v=SMBkImDWOyQ



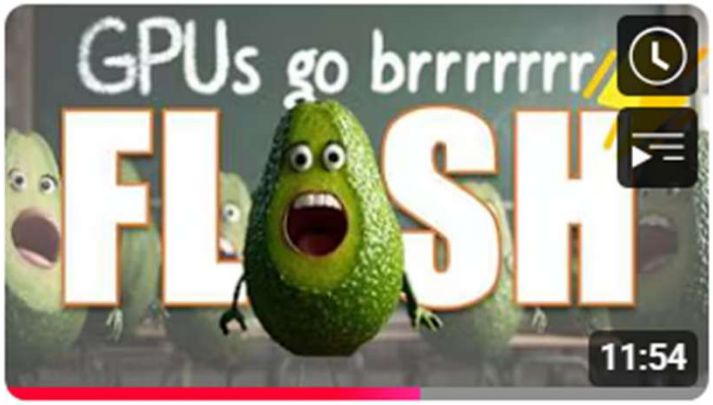

How Rotary Position Embedding Supercharges Modern LLMs...

Jia-Bin Huang • 16K views

Positional information is critical in transformers' understanding of sequences and their ability to generalize beyond training context...

# FlashAttention (from Prof. Jia-Bin Huang)

https://www.youtube.com/watch?v=gBMO1JZav44




How FlashAttention Accelerates Generative AI Revolution

Jia-Bin Huang • 16K views

FlashAttention is an IO-aware algorithm for computing attention used in Transformers. It's fast, memory-efficient, and exact. It has...